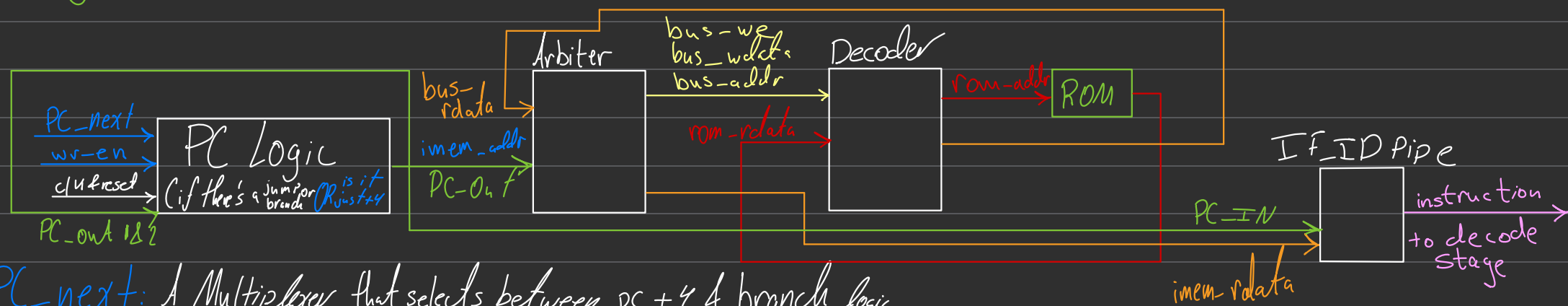


CPU Architecture 1.0

Stage 1: Fetching



PC_next: A Multiplexer that selects between $pc + 4$ & branch logic
wren: turns off the pc if either a structural or a PC stall is high

PC_out @ Arbiter: the address of the instruction

PC_out 122 @ PC Pipes: to the PC just gives it current state to +4, it also passes through all registers to calculate branching logic

bus-addr: PC's output aka instruction address & in fetching it's in the ROM's address range.
bus-wdata & bus-we = 0 in Fetch

*** Now the decoder "selects" the ROM ***

rom-addr = bus-addr: when the decoder selects ROM the `rom-addr` gets passed to the ROM and then given to the decoder through `rom-rdata`.
 $ls = bus-addr$

bus-rdata: is now populated with the instructions from the ROM equal to `rom-rdata`, the arbiter now sends it to the IF_ID pipe through `imem-rdata`